KAIS ALHASAN

M.Sc. Mechanical Engineering student, Design Automation & Product Development
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PROFILE

Final-year M.Sc. Mechanical Engineering student at Linköping University (Design & Product Development) who builds tools that automate engineering work. Looking for a 30-credit master's thesis in design automation, optimization or product development from January 2027.

EXPERIENCE

Väderstad AB, Junior Engineer, VJET program | *Väderstad* Jun 2025, Sep 2025

- Redesigned dozens of seed-drill components in Creo for function and manufacturability.
- Designed mounts that let extra GPS systems be fitted to the seed drills.
- Helped standardize components across product lines to reduce part variants.

Scania, Summer Intern, SNDP Processes & Methods (procurement) | *Södertälje* Jun 2024, Aug 2024

- Revised process documents for supplier collaboration; the procurement department implemented them.
- Ran a gap analysis of Scania's Automotive Process Manuals against the company's future goals.

Scania, Production Intern, SIP program | *Södertälje* Summers 2022 & 2023

- Built the tool register for 6-cylinder and V8 engine assembly, with cost and lifespan data for V8 tools.
- Linked each assembly process step to its tools and updated quality documents.
- Presented the gap between current state and Scania P&L's 2025 vision to the management team.

Other: Alarm operator, Verisure (2025) · Lab technician, SGS (2024-25) · Assistant, Mälarstrandens Bostäder (2021-24) · Caregiver, Linköping and Arboga (2019-24)

PROJECTS

Factory Layout Planner | *C#, .NET, Unity* 2026

- Plans a production line in checked steps and shows it in 3D; routed 97-99 % of conveyors in 2,700 test layouts.

Cable Routing with Reinforcement Learning | *Unity ML-Agents, C#, Python* 2025, 2026

- Agents that route cables through 3D CAD geometry; solve 85 of 90 unseen engine-bay routes (previous version 50).

Car Deal Finder | *Python, SQL, Power BI* 2026

- Values used cars against similar listings with regression on a SQLite star schema and ranks them by total cost.

Driving-Test Slot Monitor | *Python, Selenium* 2025

- Watched Trafikverket's booking site for canceled tests and emailed me; got my test weeks earlier.

EDUCATION

Linköping University, M.Sc. Mechanical Engineering (300 ECTS) | *Design & Product Development* 2021, Jun 2027 (expected)

- Selected courses (grade 3-5): Design Automation (4), Collaborative Design Optimization (5), Additive Manufacturing (5), CAD & Technical Drawing (5), Design Optimization (4), Conceptual Design (4).
- Bachelor's thesis (2024): kinematic and strength analysis of a dynamic speed bump.

Linköping University & Stockholm University, B.Sc. Business Administration | *parallel studies* 2025, Jun 2027 (expected)

LEADERSHIP

- Wrestling coach, IFK Linköpings Brottarklubb (2024, present): weekly training for children aged 6-12.
- Eldsjäl 2.0, RF-SISU & Folkspel (2025, 2026): leadership program; my school project brought 141 children to wrestling.

SKILLS

CAD: CATIA V5, Creo Parametric, SolidWorks, Ansys Discovery
Programming: Python, C#, VB.NET, C++, MATLAB
ML & data: Unity ML-Agents, ONNX, SQL, Power BI, Excel, Git
Languages: Swedish (fluent), English (fluent), Arabic (native)